

| Command                        | Purpose                                   | Example                                        |
|--------------------------------|-------------------------------------------|------------------------------------------------|
| git config --global user.name  | Set Git username                          | git config --global user.name "Sai Charan"     |
| git config --global user.email | Set Git email                             | git config --global user.email "abc@gmail.com" |
| git config --list              | Display current Git configuration         | git config --list                              |
| git init                       | Initialize a new Git repository           | git init                                       |
| git clone <url>                | Clone an existing repository              | git clone https://github.com/user/repo.git     |
| git status                     | Show the current status of the repository | git status                                     |
| git add <file>                 | Stage a specific file                     | git add index.html                             |
| git add .                      | Stage all modified files                  | git add .                                      |
| git add *.java                 | Stage all Java files                      | git add *.java                                 |
| git commit -m "message"        | Commit staged changes with a message      | git commit -m "Added login module"             |
| git log                        | View complete commit history              | git log                                        |
| git log --oneline              | Display commit history in one-line format | git log --oneline                              |
| git log --graph                | Display commit history as a graph         | git log --graph                                |
| git diff                       | Show unstaged changes                     | git diff                                       |
| git diff --staged              | Show staged changes                       | git diff --staged                              |
| git branch                     | List all local branches                   | git branch                                     |
| git branch <branch-name>       | Create a new branch                       | git branch feature-login                       |
| git checkout <branch>          | Switch to another branch                  | git checkout develop                           |
| git checkout -b <branch>       | Create and switch to a new branch         | git checkout -b feature-login                  |
| git switch <branch>            | Switch branches (new syntax)              | git switch main                                |
| git switch -c <branch>         | Create and switch to a new branch         | git switch -c feature-auth                     |
| git merge <branch>             | Merge a branch into the current branch    | git merge feature-login                        |
| git branch -d <branch>         | Delete a merged branch                    | git branch -d feature-login                    |
| git remote -v                  | View configured remote repositories       | git remote -v                                  |

|                             |                                                     |                                                               |
|-----------------------------|-----------------------------------------------------|---------------------------------------------------------------|
| git remote add origin <url> | Add a remote repository                             | git remote add origin<br>https://github.com/user/repo.<br>git |
| git push origin <branch>    | Push commits to a remote branch                     | git push origin main                                          |
| git push -u origin <branch> | Push and set upstream branch                        | git push -u origin main                                       |
| git pull origin <branch>    | Pull latest changes from remote                     | git pull origin main                                          |
| git fetch                   | Download remote changes without merging             | git fetch                                                     |
| git stash                   | Temporarily save uncommitted changes                | git stash                                                     |
| git stash pop               | Restore the most recent stashed changes             | git stash pop                                                 |
| git stash list              | View all stashed changes                            | git stash list                                                |
| git restore <file>          | Restore a modified file                             | git restore index.html                                        |
| git restore --staged <file> | Unstage a staged file                               | git restore --staged<br>index.html                            |
| git rm <file>               | Remove a file from Git and the working directory    | git rm test.java                                              |
| git mv <old> <new>          | Rename or move a file                               | git mv old.java new.java                                      |
| git tag                     | List all tags                                       | git tag                                                       |
| git tag <tag-name>          | Create a new tag                                    | git tag v1.0                                                  |
| git show <commit/tag>       | Show details of a commit or tag                     | git show HEAD                                                 |
| git reset --soft HEAD~1     | Undo the last commit but keep changes staged        | git reset --soft HEAD~1                                       |
| git reset --hard HEAD~1     | Undo the last commit and discard changes            | git reset --hard HEAD~1                                       |
| git revert <commit-id>      | Create a new commit that reverses a previous commit | git revert a1b2c3d                                            |
| git clean -f                | Remove untracked files                              | git clean -f                                                  |
| git clean -fd               | Remove untracked files and directories              | git clean -fd                                                 |
| git help <command>          | Display help for a Git command                      | git help commit                                               |